Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
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Longest Common Subsequence (LCS) di 3 sequenze

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Problema LCS(X,Y,W)

Date tre sequenze

$X = \langle x_1, x_2, \ldots, x_m \rangle$, $Y = \langle y_1, y_2, \ldots, y_n \rangle$, $W = \langle w_1, w_2, \ldots, w_l \rangle$

trovare la più lunga sottosequenza comune di X, Y e W

$(m,n,l) \rightarrow$ dimensione del problema

Sottoproblemi

Sottoproblema di dimensione (i, j, h)

Trovare LCS(X_i, Y_j, W_h), che è la LCS dei prefissi X_i, Y_j e W_h

$i \in \{0, 1, \ldots, m\}$

$j \in \{0, 1, \ldots, n\}$

$h \in \{0, 1, \ldots, l\}$

Numero totale di sottoproblemi: $(m+1)(n+1)\ (l+1)$

LCS(X,Y,W) $\equiv$ sottoproblema di dimensione (m,n,l)

Sottoproblemi

Sottoproblema di dimensione (i, j, h)

Trovare LCS(X_i, Y_j, W_h), che è la LCS dei prefissi X_i, Y_j e W_h

$i \in \{0, 1, \ldots, m\}$

$j \in \{0, 1, \ldots, n\}$

$h \in \{0, 1, \ldots, l\}$

$c_{i,j,h} = |LCS(X_i, Y_j, W_h)|$

Numero totale di sottoproblemi: (m+1)(n+1) (l+1)

LCS(X,Y,W) ≡ sottoproblema di dimensione (m,n,l)

Sottoproblemi

Sottoproblema di dimensione (i, j, h)

Trovare LCS(X_i, Y_j, W_h), che è la LCS dei prefissi X_i, Y_j e W_h

$i \in \{0, 1, \ldots, m\}$

$j \in \{0, 1, \ldots, n\}$

$h \in \{0, 1, \ldots, l\}$

$c_{i,j,h} = |LCS(X_i, Y_j, W_h)|$

Numero totale di sottoproblemi: $(m+1)(n+1)\ (l+1)$

Valore ottimo $\equiv c_{m,n,l}$

Equazione di ricorrenza

$$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$$
$$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$$
$$W_h = \langle w_1, w_2, \ldots, w_{h-1}, w_h \rangle$$

$i = 0 \lor j = 0 \lor h = 0$ (CASI BASE)

$LCS(X_i, Y_j, W_h) = \langle\rangle$

$\mathbf{c_{i,j,h} = 0}$

$i > 0 \land j > 0 \land h > 0$ (PASSO RICORSIVO)

$x_i = y_j \land y_j = w_h$

$\Rightarrow LCS(X_i, Y_j, W_h) = LCS(X_{i-1}, Y_{j-1}, W_{h-1}) + \langle x_i \rangle$

$\Rightarrow \mathbf{c_{i,j,h} = c_{i-1,j-1,h-1} + 1}$

$x_i \neq y_j \lor y_j \neq w_h$

$\Rightarrow LCS(X_i, Y_j, W_h) = \max\{LCS(X_{i-1}, Y_j, W_h), LCS(X_i, Y_{j-1}, W_h), LCS(X_i, Y_j, W_{h-1})\}$

$\Rightarrow \mathbf{c_{i,j,h} = \max\{c_{i-1,j,h}, c_{i,j-1,h}, c_{i,j,h-1}\}}$

Algoritmo bottom-up (DP)

```
int calcola_ottimo_LCS_3_sequenze(X,Y,W)
    C ← matrice di dimensione (m+1)(n+1)(l+1)
    for i from 0 to m do
        for j from 0 to n do
            C[i,j,0] ← 0
    for i from 0 to m do
        for h from 0 to l do
            C[i,0,h] ← 0
    for j from 0 to n do
        for h from 0 to l do
            C[0,j,h] ← 0
    for i from 1 to m do
        for j from 1 to n do
            for h from 1 to l do
                if x_i = y_j and y_j = w_h then
                    C[i,j,h] ← C[i-1,j-1,h-1] + 1
                else
                    C[i,j,h] ← max(C[i-1,j,h],C[i,j-1,h],C[i,j,h-1])
    return C[m,n,l]
```

Algoritmo bottom-up (DP)

```
int calcola_ottimo_LCS_3_sequenze(X,Y,W)
    C ← matrice di dimensione (m+1)(n+1)(l+1)
    for i from 0 to m do
        for j from 0 to n do
            C[i,j,0] ← 0
    for i from 0 to m do
        for h from 0 to l do
            C[i,0,h] ← 0
    for j from 0 to n do
        for h from 0 to l do
            C[0,j,h] ← 0
    for i from 1 to m do
        for j from 1 to n do
            for h from 1 to l do
                if xi = yj and yj = wh then
                    C[i,j,h] ← C[i-1,j-1,h-1] + 1
                else
                    C[i,j,h] ← max(C[i-1,j,h],C[i,j-1,h],C[i,j,h-1])
    return C[m,n,l]
```

θ(mnl) in tempo/spazio

Ricostruzione di LCS(X_i,Y_j,W_h)

```
procedura Print_LCS_3_sequenze(C, X, i, j, h)
   if i > 0 and j > 0 and h > 0 then
      if xi = yj and yj = wh then
          Print_LCS_3_sequenze(C, X, i-1, j-1, h-1)
          print xi
      else
         if C[i,j,h] = C[i-1, j, h] then
             Print_LCS_3_sequenze(C, X, i-1, j, h)
         else
            if C[i,j,h] = C[i, j-1, h] then
                Print_LCS_3_sequenze(C, X, i, j-1, h)
            else
                Print_LCS_3_sequenze(C, X, i, j, h-1)
```